Hybrid Sphere Decoding with Embedded Initialization and Explicit Pruning

A paired study of leaf-cost equivalence, partial metrics, pruning margins, and sphere-decoder search paths

Thesis Analysis Note

This document develops the argument for a hybrid sphere decoder that uses the embedded zero-vector formulation to construct its initial incumbent and the explicit formulation to perform exact branch-and-bound enumeration. The argument is organized around leaf-level objective equivalence, upper-bound quality, prefix-bound tightness, and a controlled ablation study. Its purpose is to provide a rigorous source for later integration into the master thesis.

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Contents

1 Terminology and How to Read the Diagnostics **1**

1.1 Search tree, node, depth, prefix, and leaf 1

1.2 Partial metric, incumbent, radius, and pruning 2

1.3 Paired observation and paired node-count difference 2

1.4 Probability density and kernel-density estimate 3

1.5 Median, quartiles, IQR, and representative percentiles 3

1.6 Same-prefix comparison and initial-sphere membership 4

1.7 Effective offset and representation-induced prefix slack 4

2 Problem Statement and Experimental Observation **5**

2.1 Central argument of the hybrid method 5

3 Leaf-Level Equivalence of the Two Penalties **7**

3.1 Exact binary representation 7

3.2 Why leaf equivalence does not prove search equivalence 8

4 Search Mechanism and Working Hypotheses **9**

4.1 Explicit block-boundary penalty 9

4.2 Embedded quadratic representation 9

4.3 Refined prefix-slack hypothesis 9

4.4 Evidence status 10

4.5 Block-boundary prefix relation 10

5 Diagnostic Quantities and Their Interpretation **12**

5.1 Paired node difference 12

5.2 Visited, pruned, and surviving nodes by depth 12

5.3 Pruning margin $R - P_i$ 13

5.4 Dynamic incumbent radius 13

5.5 Node-level trace 13

6 Reproducible Experimental Design **14**

7 Proposed Hybrid Sphere Decoder **15**

7.1 Separation of upper-bound generation and lower-bound search 15

7.2 Algorithm definition 15
7.3 Discovery and proof effort 16
7.4 Controlled ablation design 16

8 Current Results **18**
8.1 Population-level comparison 18
8.2 Depth-resolved behavior 19
8.3 Representative median case 19
8.4 Representative-case definition 21
8.5 Complete same-prefix enumeration 21
8.6 Hybrid and oracle ablation results 23
8.6.1 Evidence for the embedded upper-bound advantage 24
8.6.2 Evidence for the explicit proof advantage 25
8.6.3 Interpretation of the hybrid improvement 25

9 Remaining Analysis **26**

10 Reproducibility Assets **27**

1 Terminology and How to Read the Diagnostics

This chapter defines the search-specific terms used in the report. They are introduced here because several familiar statistical words have a precise meaning in the present SDA experiment.

1.1 Search tree, node, depth, prefix, and leaf

A *search tree* is a representation of all binary switching sequences. At each level, the search chooses either zero or one for the next element of $\boldsymbol{U}$. A *node* is one such attempted choice together with the choices already made before it. Visiting a node means that the solver has formed this partial candidate and calculated its partial metric; it does not mean that the candidate is accepted or optimal.

The *depth* i is the number of binary variables currently fixed. The implemented forward search begins at $i = 1$ and ends at $i = 15$. The vector of the first i fixed decisions is called a *prefix*, or partial switching sequence:

$$\boldsymbol{U}_{1:i} = [U_1, U_2, \ldots, U_i]^{\mathsf{T}}. \tag{1.1}$$

For example, if a complete candidate begins with

$$\boldsymbol{U} = [0, 0, 1,\ 1, 0, 1, \ldots]^{\mathsf{T}}, \tag{1.2}$$

then its prefixes at depths 1, 3, and 6 are 0, 001, and 001101, respectively. A depth-three prefix is the first complete three-phase switching vector. Complete switching-vector boundaries therefore occur at $i = 3, 6, 9, 12, 15$. At depth i there are 2^i possible prefixes.

A *leaf* is a node at depth 15, where every element of $\boldsymbol{U}$ is known. Only a leaf represents a complete admissible switching sequence and has a full objective value. An internal node represents only a prefix and is evaluated using a lower, partial cost.

1.2 Partial metric, incumbent, radius, and pruning

The *partial metric* P_i is the cost accumulated from the information available at depth i. It is a lower bound used to decide whether completing the prefix can still be worthwhile. It is not generally the final physical cost of a switching sequence.

The best complete sequence found so far is the *incumbent*. Its cost is the current *sphere radius* R. "Radius" is inherited from the geometric interpretation of sphere decoding; in the program, it is a scalar cost threshold rather than a physical length. The initial incumbent is provided by the best of the Round, Shift, and Babai candidates. Whenever the search finds a better leaf, that leaf becomes the new incumbent and R decreases. The term *dynamic radius* emphasizes that R can change many times during one SDA call.

Pruning means terminating a branch because its partial metric already exceeds the incumbent:

$$P_i > R. \tag{1.3}$$

The descendants of that prefix are then not visited. A *surviving node* is a tested node with $P_i \le R$ that may be expanded to the next depth.

The *pruning margin* is

$$M_i = R - P_i. \tag{1.4}$$

It can be read as the remaining cost allowance of the tested prefix:

- $M_i > 0$: the prefix is still inside the current sphere and survives;
- $M_i = 0$: the prefix lies on the pruning boundary;
- $M_i < 0$: the prefix is outside the sphere and is pruned.

A large positive margin does not mean that the prefix is close to optimal. It only means that the available lower bound is not yet large enough to reject it.

1.3 Paired observation and paired node-count difference

Two observations are *paired* when they describe the same operating condition and the same controller transaction. Thus, the embedded and explicit node counts at time t_k form one pair. The paired node-count difference is

$$\Delta N_k = N_{\mathrm{emb},k} - N_{\mathrm{exp},k}. \tag{1.5}$$

A positive value means that the embedded formulation visited more nodes for that same physical case. Pairing prevents a heavily loaded embedded sample from being compared accidentally with a lightly loaded explicit sample.

1.4 Probability density and kernel-density estimate

A histogram divides node differences into bins. A *kernel-density estimate* (KDE) is a smooth version of the same distribution. Each measured ΔN_k contributes a small smooth kernel, and their sum forms the plotted curve. The vertical axis is *probability density*, not the probability of one exact node difference. The area under the complete curve equals one:

$$\int_{-\infty}^{\infty} \widehat{f}_h(x)\,\mathrm{d}x = 1. \tag{1.6}$$

The probability that ΔN lies in an interval is the area under the curve over that interval. Consequently, a vertical value such as 1.4×10^{-3} should not be interpreted as 0.14% of the samples. Its unit is approximately inverse nodes, and it becomes a probability only after integration over a node-count interval.

1.5 Median, quartiles, IQR, and representative percentiles

The *median*, also denoted Q_{50}, is the value below which half the observations lie. The first and third quartiles, Q_{25} and Q_{75}, enclose the middle half of the observations. The *interquartile range* (IQR) is the interval

$$\mathrm{IQR} = [Q_{25}, Q_{75}]. \tag{1.7}$$

In the figures, a solid line commonly shows the median and a shaded band shows the IQR. A wide band indicates strong path-to-path variation.

The labels Q_5, Q_{50}, and Q_{95} used for representative transactions refer to percentiles of the paired node difference ΔN: a weak difference case, a typical case, and a pronounced difference case. They do not refer to motor torque, confidence intervals, or prediction horizons.

1.6 Same-prefix comparison and initial-sphere membership

The actual embedded and explicit searches visit different nodes in different orders. Their 100th visited nodes therefore need not represent the same partial switching sequence. A *same-prefix comparison* instead fixes one specific pair $(i, \boldsymbol{U}_{1:i})$ and evaluates that identical prefix under both cost representations. Enumerating all 2^i prefixes at each depth produces a comparison that is independent of the actual traversal order.

A prefix is described as *inside the initial sphere* when its partial metric does not exceed the radius before any search-generated radius update:

$$P_i \leq R_0. \tag{1.8}$$

This is a static diagnostic. It does not predict the complete real search on its own because the actual radius may shrink after an improved leaf is found.

1.7 Effective offset and representation-induced prefix slack

For an identical prefix, the effective accumulated difference is

$$C_i^{\mathrm{eff}} = P_i^{\mathrm{emb}} - P_i^{\mathrm{exp}}. \tag{1.9}$$

It answers: "At this depth, how much more partial metric has the embedded representation accumulated for the same binary choices?" The representation-induced prefix slack is

$$S_i = 6250 - C_i^{\mathrm{eff}}. \tag{1.10}$$

A positive S_i means that the full leaf offset 6250 has not yet appeared in the difference between the two partial metrics. It does not automatically mean that the embedded node survives, because each formulation can have a different and dynamically changing radius.

2 Problem Statement and Experimental Observation

The long-horizon finite-control-set model predictive controller uses a binary sphere decoder to determine a switching sequence

$$\boldsymbol{U} = [S_{a,1}, S_{b,1}, S_{c,1}, \ldots, S_{a,N_p}, S_{b,N_p}, S_{c,N_p}]^{\mathsf{T}}, \qquad \boldsymbol{U} \in \{0,1\}^{3N_p}, \tag{2.1}$$

with $N_p = 5$ and hence 15 binary decision variables. Two mathematically equivalent treatments of the zero vectors 000 and 111 were implemented. The first adds a discrete penalty when a complete three-switch block is known. The second incorporates an equivalent quadratic expression into the Hessian before its Cholesky factorization.

Paired simulations at 10, 50, and 90 Nm produced the following initially surprising observations.

1. The two formulations returned identical optimal switching sequences in all paired samples considered.
2. The embedded formulation reported an optimal cost approximately 6250 larger than the explicit formulation at every transaction.
3. The embedded formulation visited substantially more search nodes.
4. Babai initialization was the best of the tested initializers in both formulations and dominated especially strongly in the embedded formulation.

These results separate control equivalence from computational equivalence. An identical minimizer at the leaves of a search tree does not imply identical partial metrics or identical pruning decisions inside the tree.

2.1 Central argument of the hybrid method

The analysis is organized around three claims.

1. The explicit and embedded objectives are equivalent on complete binary sequences up to a constant. A feasible sequence obtained from one representation can therefore initialize the other without changing the optimizer.

2. The embedded geometry includes the zero-vector preference when the Babai point is constructed and consequently provides a substantially better initial upper bound than the native explicit initializer in the investigated cases.

3. The explicit formulation applies the exact penalty at complete three-switch block boundaries and supplies tighter normalized prefix bounds at these depths. It therefore requires fewer nodes to certify optimality.

The proposed hybrid uses the embedded formulation only for incumbent generation and then evaluates that incumbent under the exact explicit objective before commencing explicit SDA enumeration.

3 Leaf-Level Equivalence of the Two Penalties

3.1 Exact binary representation

For one three-phase switching vector define

$$m = S_a + S_b + S_c, \qquad m \in \{0,1,2,3\}. \tag{3.1}$$

The vector is a zero vector exactly when $m = 0$ or $m = 3$. The explicit cost is

$$J_{\mathrm{z,exp}}(m) = \begin{cases} \lambda_z, & m \in \{0,3\}, \\ 0, & m \in \{1,2\}. \end{cases} \tag{3.2}$$

On the binary domain this logic is represented exactly by

$$J_{\mathrm{z,exp}}(m) = \frac{\lambda_z}{2}(m-1)(m-2). \tag{3.3}$$

The centered quadratic used by the embedded formulation is

$$J_{\mathrm{z,emb}}(m) = \frac{\lambda_z}{2}(m-1.5)^2. \tag{3.4}$$

Consequently,

$$\boxed{J_{\mathrm{z,emb}}(m) = J_{\mathrm{z,exp}}(m) + \frac{\lambda_z}{8}.} \tag{3.5}$$

For $\lambda_z = 10{,}000$, the explicit costs of zero and active vectors are 10,000 and zero, whereas the embedded costs are 11,250 and 1,250. Both therefore retain the same zero-to-active cost difference of 10,000. The embedded representation merely adds 1250 to every complete switching vector. Over $N_p = 5$ steps,

$$C = \frac{N_p \lambda_z}{8} = 6250. \tag{3.6}$$

For every complete binary sequence,

$$J_{\mathrm{emb}}(\boldsymbol{U}) = J_{\mathrm{exp}}(\boldsymbol{U}) + C, \tag{3.7}$$

and therefore both formulations have the same minimizer.

Proposition 3.1 (Leaf-level objective equivalence)**.** *For $\boldsymbol{U} \in \{0,1\}^{3N_p}$, the embedded and explicit objectives satisfy*

$$J_{\mathrm{emb}}(\boldsymbol{U}) = J_{\mathrm{exp}}(\boldsymbol{U}) + \frac{N_p \lambda_z}{8}. \tag{3.8}$$

Consequently, their complete-sequence minimizer sets are identical.

Proof. Equation (3.5) holds independently for every prediction block. Summing it over N_p blocks produces a sequence-independent constant. Adding a constant to all feasible objective values preserves every pairwise ordering and therefore preserves the set of minimizers. □

3.2 Why leaf equivalence does not prove search equivalence

The SDA does not wait until all 15 variables are known. With the lower triangular factor used by the implemented forward search, its quadratic partial metric at depth i is

$$P_i^{\mathrm{quad}} = \sum_{k=1}^{i} \left((HU_{\mathrm{unc}})_k - \sum_{j=1}^{k} H_{kj} U_j \right)^2. \tag{3.9}$$

A branch is pruned when its complete partial metric exceeds the current incumbent radius R:

$$P_i > R. \tag{3.10}$$

Equation (3.6) constrains the leaf cost at $i = 15$, but it does not imply that partial metrics at $i < 15$ differ by a fixed or uniformly distributed fraction of 6250.

4 Search Mechanism and Working Hypotheses

4.1 Explicit block-boundary penalty

The production search proceeds from $i = 1$ to $i = 15$. At $i = 3, 6, 9, 12, 15$, a complete three-phase vector is available. The explicit formulation then evaluates the zero-vector condition and, if necessary, adds λ_z immediately. This creates a semantic alignment between the physical event “a zero vector has been completed” and a discrete jump in the partial search metric.

4.2 Embedded quadratic representation

Using $S_x^2 = S_x$ for binary variables, the embedded block cost expands to

$$J_{\mathrm{z,emb}} = \frac{9\lambda_z}{8} - \lambda_z(S_a + S_b + S_c) + \lambda_z(S_aS_b + S_aS_c + S_bS_c). \tag{4.1}$$

It is therefore not composed of three independent one-third penalties. Its final classification is produced jointly by linear and pairwise terms. After the complete Hessian is factorized,

$$\boldsymbol{W} = \boldsymbol{H}\boldsymbol{H}^{\mathsf{T}}, \tag{4.2}$$

the original block meaning is generally redistributed among several squared residuals. The individual residual increment at one search level is thus not itself interpretable as a fraction of the zero-vector penalty.

4.3 Refined prefix-slack hypothesis

For the same binary prefix, define the effective accumulated difference

$$C_i^{\mathrm{eff}} = P_i^{\mathrm{emb}} - P_i^{\mathrm{exp}} \tag{4.3}$$

and the remaining representation-induced prefix slack

$$S_i = C - C_i^{\mathrm{eff}}. \tag{4.4}$$

If the two initial radii differ by C, the pruning margins satisfy

$$M_i^{\mathrm{exp}} = R_{\mathrm{exp}} - P_i^{\mathrm{exp}}, \tag{4.5}$$

$$M_i^{\mathrm{emb}} = R_{\mathrm{emb}} - P_i^{\mathrm{emb}}, \tag{4.6}$$

$$M_i^{\mathrm{emb}} - M_i^{\mathrm{exp}} = S_i. \tag{4.7}$$

A positive S_i means that the embedded representation leaves more room before the same prefix reaches its pruning boundary. At a complete leaf, $C_{15}^{\mathrm{eff}} = C$ and $S_{15} = 0$ are expected. At intermediate depths, neither quantity is assumed to be uniform or path independent.

The constant 6250 is not random noise. It is a deterministic common offset. Nor is a larger numerical radius alone sufficient to explain additional nodes: adding the same constant consistently to every bound and every radius would leave all pruning decisions unchanged. The proposed mechanism is the possible mismatch between the offset already present in a complete incumbent and the amount represented in a shallow partial metric.

4.4 Evidence status

Three statements must remain distinct.

1. **Proved:** the complete binary objectives differ by the constant $C = 6250$.
2. **Observed:** optimal switching sequences coincide while the embedded search visits more nodes.
3. **Hypothesized mechanism:** factorization changes the tightness and depth distribution of prefix bounds, delaying pruning for many prefixes.

The third statement requires same-prefix analysis and must not be presented as a proven causal conclusion solely from row norms of H.

4.5 Block-boundary prefix relation

Under the implemented forward variable ordering, a depth $i = 3q$ prefix contains exactly q complete switching vectors. The lower-triangular search factor makes the first $3q$ residuals

depend only on these fixed variables. The per-block equivalence in equation (3.5) can therefore be applied to the truncated prefix objective.

Proposition 4.1 (Completed-block offset)**.** *At the completed switching-vector depths $i = 3q$,*

$$P_{3q}^{\mathrm{emb}} - P_{3q}^{\mathrm{exp}} = q\frac{\lambda_z}{8}. \tag{4.8}$$

For the normalized embedded objective $\widetilde{J}_{\mathrm{emb}} = J_{\mathrm{emb}} - N_p\lambda_z/8$,

$$P_{3q}^{\mathrm{exp}} - widetildeP_{3q}^{\mathrm{emb}} = (N_p - q)\frac{\lambda_z}{8} > 0, \qquad q < N_p. \tag{4.9}$$

Proof. Each of the q completed blocks contributes the constant $\lambda_z/8$ between the two representations, while the remaining $N_p - q$ block constants belong to unresolved variables and are absent from the current prefix metric. Subtracting the complete leaf constant $N_p\lambda_z/8$ from the embedded metric yields equation (4.9). □

For $N_p = 5$ and $\lambda_z = 10{,}000$, the measured offsets at depths $3, 6, 9, 12, 15$ are respectively $1250, 2500, 3750, 5000, 6250$ for every enumerated prefix. Between these block boundaries the factorized cross terms make the offset path dependent; no universal ordering is claimed there.

5 Diagnostic Quantities and Their Interpretation

5.1 Paired node difference

For transaction k, paired by operating condition and controller time, define

$$\Delta N_k = N_{\mathrm{emb},k} - N_{\mathrm{exp},k}. \tag{5.1}$$

Thus $\Delta N_k > 0$ means that the embedded solver visited more nodes for the same physical transaction. Pairing is essential: two unpaired node-count samples may correspond to different motor states and are not a controlled comparison.

The plotted probability-density curves are kernel-density estimates (KDEs):

$$\widehat{f}_h(x) = \frac{1}{nh} \sum_{k=1}^{n} K\left(\frac{x - \Delta N_k}{h} \right), \tag{5.2}$$

where K is a normalized smoothing kernel and h is its bandwidth. The vertical scale is probability density, not sample probability: the area under each curve equals one. A curve concentrated to the right of zero indicates that positive node differences dominate the paired population.

5.2 Visited, pruned, and surviving nodes by depth

Let N_i^{vis} be the number of candidates evaluated at depth i, N_i^{pr} the number rejected by (3.10), and N_i^{sur} the number allowed to descend. The depth-wise pruning rate is

$$p_i^{\mathrm{prune}} = \frac{N_i^{\mathrm{pr}}}{N_i^{\mathrm{vis}}}. \tag{5.3}$$

Peaks at $i = 3, 6, 9, 12, 15$ are of particular interest because these are the three-switch block boundaries.

5.3 Pruning margin $R - P_i$

For every visited node, the pruning margin is defined as

$$M_i = R - P_i. \tag{5.4}$$

Here R is the incumbent radius immediately before the node is tested, not necessarily the initial radius. Its interpretation is:

$$M_i > 0 : \text{node remains inside the sphere}, \qquad M_i < 0 : \text{node is pruned}. \tag{5.5}$$

The median and interquartile range summarize the path-dependent distribution at each depth. A large positive margin does not mean that a candidate is good; it only means that the current lower bound has not yet excluded it.

5.4 Dynamic incumbent radius

The incumbent radius can shrink whenever a better complete leaf is found. Plotting R against visited-node index distinguishes two effects:

1. initialization quality, which determines the first radius; and
2. discovery order, which determines how quickly subsequent leaves tighten the radius.

Consequently, comparing initial radii alone is insufficient.

5.5 Node-level trace

The independent profiler records the search depth, binary candidate, encoded prefix, incoming metric, squared residual, zero-vector increment, dead-time increment, complete partial metric, radius, pruning margin, pruning decision, leaf flag, and radius update. Zero-vector and dead-time increments are stored separately so that a block-boundary rejection can be assigned to its actual cause.

6 Reproducible Experimental Design

The SDI archive contains 40,000 SDA transactions for each run. The analysis extracts all inputs and reference outputs and replays them with an independent MATLAB implementation of the production search. The replay is accepted only when J_{opt}, the node count, and $\boldsymbol{U}_{\mathrm{opt}}$ match the logged reference.

Two complementary analyses are required.

1. **Actual-search replay** records the nodes genuinely visited by Schnorr–Euchner enumeration, including dynamic radius updates.
2. **Complete same-prefix enumeration** evaluates both formulations for the same (i, prefix) pairs. It avoids the selection bias that would result from comparing the kth visited node of two different search paths.

Representative transactions are selected reproducibly from the distribution of ΔN, rather than chosen visually. For each torque condition, the cases nearest the 5th, 50th, and 95th percentiles are used. The 50th percentile represents typical behavior, the 95th percentile represents a pronounced complexity increase, and the 5th percentile acts as a weak-difference control.

7 Proposed Hybrid Sphere Decoder

7.1 Separation of upper-bound generation and lower-bound search

The embedded and explicit formulations are assigned different roles. The embedded factor generates a high-quality feasible sequence,

$$\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}} = \mathrm{Babai}(\boldsymbol{H}_{\mathrm{emb}}, \boldsymbol{U}_{\mathrm{unc,emb}}), \tag{7.1}$$

because its nearest-plane geometry already includes the zero-vector weight. This sequence is then evaluated under the explicit objective:

$$R_{0,\mathrm{hyb}} = J_{\mathrm{exp}}(\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}). \tag{7.2}$$

The explicit factor and exact block-boundary penalties subsequently perform the branch-and-bound search. Thus, the embedded formulation supplies the initial upper bound, whereas the explicit formulation supplies the recursive prefix lower bounds.

7.2 Algorithm definition

For each controller transaction the hybrid procedure is:

1. construct the embedded factor and unconstrained center;
2. perform the forward Babai pass to obtain $\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}$;
3. evaluate this same binary sequence under the explicit quadratic, zero-vector, and dead-time objective;
4. initialize the explicit incumbent with $(\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}, R_{0,\mathrm{hyb}})$;
5. run the original explicit SDA, adding the branch penalties at $i = 3, 6, 9, 12, 15$;

6. return the best complete sequence found by the explicit search.

Proposition 7.1 (Global optimality of the hybrid solver)**.** *Assume that the explicit partial metric is a valid lower bound for every completion of its prefix and that the SDA terminates without reaching its node limit. Initializing it with the embedded Babai sequence according to equation* (7.2) *does not change the globally optimal solution.*

Proof. The embedded Babai sequence is binary and complete, hence feasible for the explicit problem. Its explicit cost is therefore an upper bound:

$$R_{0,\mathrm{hyb}} = J_{\mathrm{exp}}(\boldsymbol{U}_{\mathrm{B}}^{\mathrm{emb}}) \geq J_{\mathrm{exp}}^{\star}. \tag{7.3}$$

The explicit SDA prunes only a prefix whose valid lower bound exceeds the best known feasible cost. Such a prefix cannot contain a better leaf. Changing the feasible sequence used to establish the first upper bound can change the traversal and node count, but cannot remove the global optimum. □

7.3 Discovery and proof effort

Let $N_{\text{first-opt}}$ be the visited-node index at which the incumbent first reaches the known offline optimum. Define

$$N_{\mathrm{discovery}} = N_{\text{first-opt}}, \qquad N_{\mathrm{proof}} = N_{\mathrm{total}} - N_{\text{first-opt}}. \tag{7.4}$$

If the initializer is already optimal, $N_{\mathrm{discovery}} = 0$. Remaining nodes quantify the work performed after the optimum first becomes available. This is a diagnostic accounting convention: nodes visited before discovery may also contribute to the final proof and the terms are not independent algorithms.

7.4 Controlled ablation design

Five configurations are evaluated on identical transactions:

Native explicit: original initializer and explicit search;

Native embedded: embedded initializer and embedded search;

Hybrid: embedded Babai sequence and explicit search;

Oracle explicit: known optimal sequence supplied to explicit search;

Oracle embedded: known optimal sequence supplied to embedded search.

The oracle configurations are diagnostic rather than implementable controllers. They remove initialization error and measure the work each representation requires to certify an already known optimum. Every tested configuration is checked against the logged optimum.

8 Current Results

8.1 Population-level comparison

The full paired data show that the node-difference distributions are located predominantly to the right of zero, while the leaf-cost difference agrees with 6250 to numerical precision.

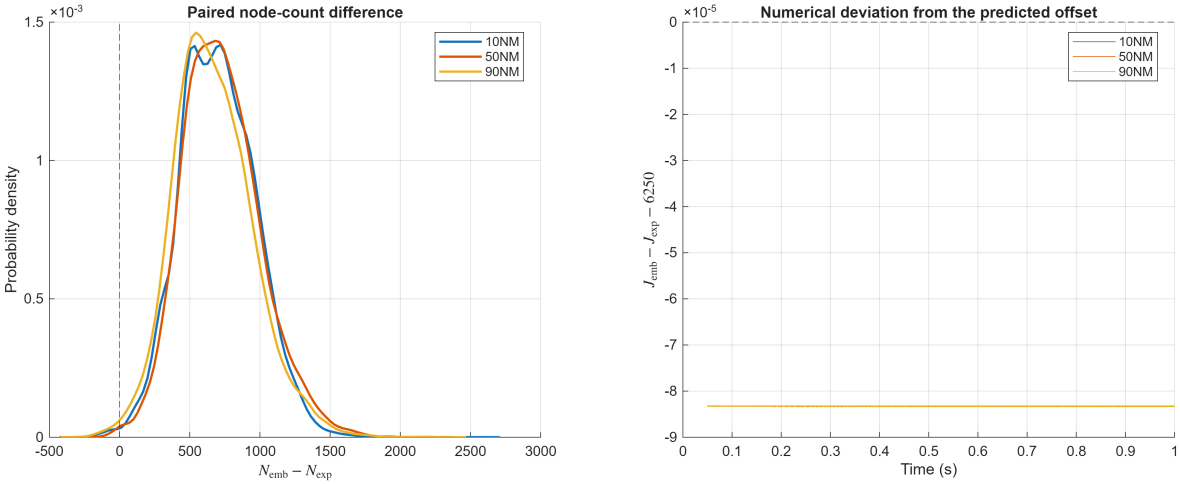


Figure 8.1: Left: kernel-density estimate of the paired node-count difference $\Delta N = N_{\text{emb}} - N_{\text{exp}}$; positive values mean that the embedded solver visited more nodes. The area below each curve equals one, whereas its height is a density rather than a point probability. Right: numerical deviation of the measured leaf-cost difference from the predicted constant 6250.

Independent replay of 500 uniformly distributed transactions per run produced the mean node counts in table 8.1. Every replayed transaction matched the logged optimum, node count, and switching sequence.

Table 8.1: Mean visited nodes in the independently replayed sample.

Torque	Embedded H	Explicit penalty	Difference
10 Nm	1151.87	479.55	672.32
50 Nm	1177.64	474.69	702.95
90 Nm	1095.46	462.34	633.12

8.2 Depth-resolved behavior

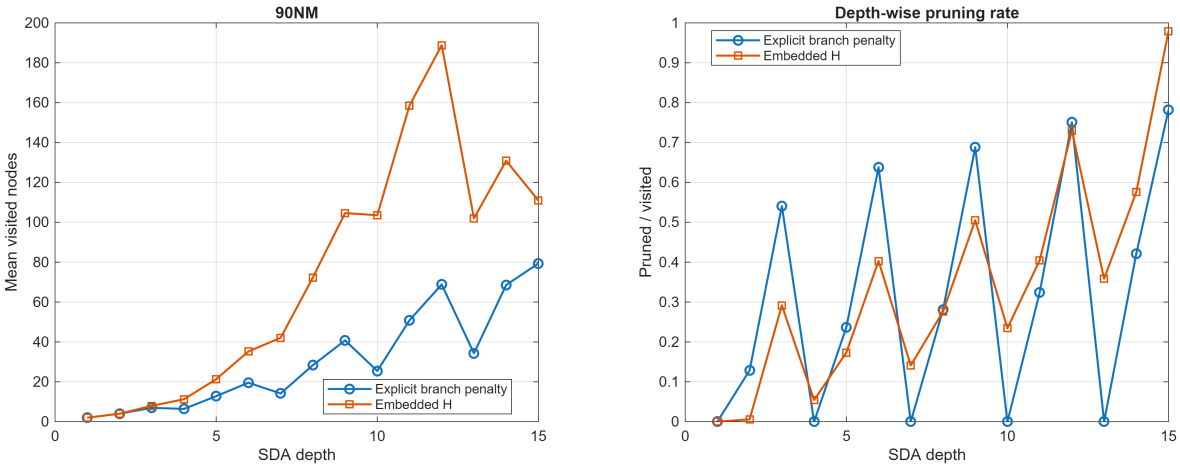


Figure 8.2: Mean visited nodes and aggregate pruning rate versus search depth at 90 Nm. The explicit formulation exhibits pronounced pruning-rate peaks at several complete three-switch block boundaries, while the embedded formulation carries more surviving candidates into deeper levels.

The same qualitative behavior appears at 10 and 50 Nm. This result supports the block-boundary-alignment hypothesis, but does not yet prove that the 6250 offset is the sole cause.

8.3 Representative median case

At 90 Nm, the median node-difference case occurs at transaction 2305 and $t = 0.057625$ s. The explicit search visits 494 nodes and the embedded search 1140 nodes, giving $\Delta N = 646$.

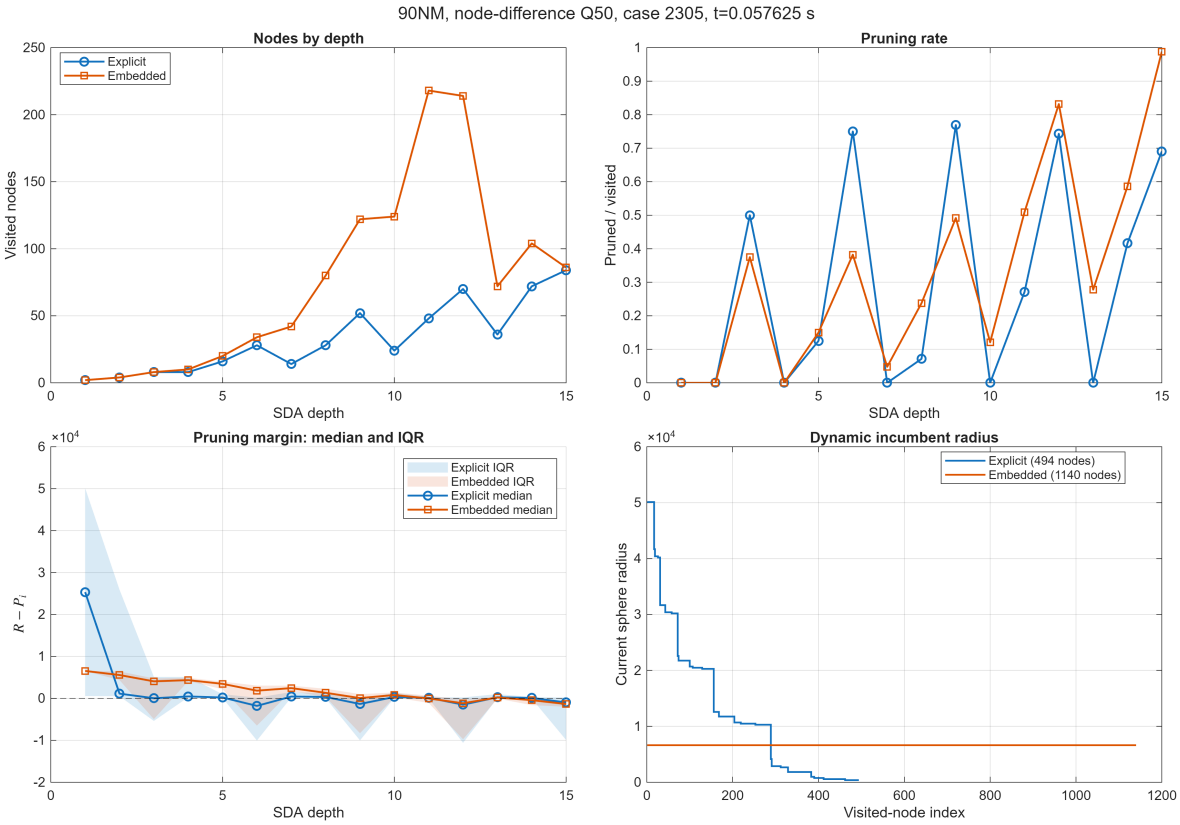


Figure 8.3: Search diagnostics for the 90 Nm median node-difference case. The lower-left panel reports $M_i = R - P_i$ using the radius immediately before each tested node: positive values survive and negative values are pruned. Solid lines are medians and shaded regions contain the middle 50% of observations (IQR). The lower-right panel shows how the best known complete cost, called the incumbent radius, changes with visited-node index.

The figure adds an important qualification to the initial hypothesis. In this case the embedded Babai incumbent is already close to its final value and its radius remains almost constant. The explicit formulation starts from a much larger incumbent but discovers better leaves early and rapidly contracts its radius. The embedded solver nevertheless visits more nodes, especially at intermediate depths. Hence the raw initial-radius magnitude cannot by itself explain the complexity difference. The evidence instead points to the joint effect of prefix-bound tightness, block-boundary penalties, enumeration order, and dynamic incumbent updates.

8.4 Representative-case definition

The selected (Q_5, Q_{50}, Q_{95}) node differences are respectively $(282, 692, 1148)$ at 10 Nm, $(314, 706, 1226)$ at 50 Nm, and $(244, 646, 1166)$ at 90 Nm. All corresponding figures and exact transaction indices are retained with the analysis data to prevent subjective figure selection.

8.5 Complete same-prefix enumeration

The complete prefix tree was subsequently enumerated for all nine selected transactions. At depth i, all 2^i binary prefixes were evaluated under both formulations. This removes the ambiguity of comparing differently ordered, actually visited nodes. At every complete leaf and in every selected case, the measured difference was

$$P_{15}^{\mathrm{emb}} - P_{15}^{\mathrm{exp}} = 6249.99992 \tag{8.1}$$

for all 2^{15} sequences; the tiny deviation from 6250 is numerical.

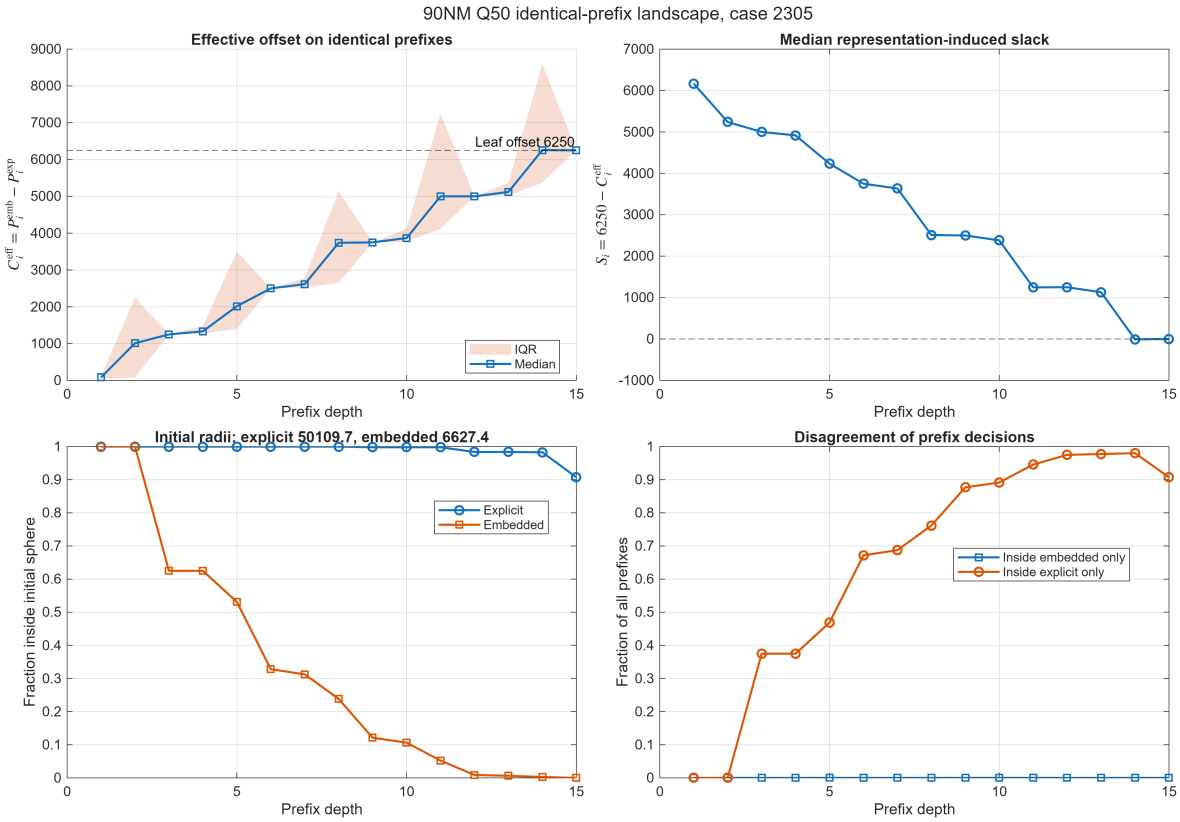


Figure 8.4: Complete same-prefix landscape for the 90 Nm median case. Top left: effective prefix offset and its interquartile range. Top right: remaining median slack. Bottom left: fraction of all prefixes inside each formulation's initial sphere. Bottom right: prefixes classified inside only one of the two initial spheres.

The median C_i^{eff} increases progressively toward 6250, and the corresponding median S_i decreases toward zero. This confirms that the leaf offset is not uniformly present in shallow partial metrics. However, the initial-radius comparison provides an important counterexample to a simple "larger embedded sphere" explanation. In this transaction,

$$R_{0,\text{exp}} = 50109.7, \qquad R_{0,\text{emb}} = 6627.4. \tag{8.2}$$

The explicit initial sphere contains almost all prefixes through most depths, whereas the embedded initial sphere is substantially tighter. A static test against the initial radii would therefore predict more aggressive embedded pruning, opposite to the actually observed node count.

The discrepancy is resolved by recognizing that the initial sphere is not the sphere used throughout the search. As shown in figure 8.3, the explicit search discovers progressively better leaves and rapidly reduces its incumbent radius. The embedded Babai incumbent begins near its final value and remains almost unchanged. Actual complexity is

consequently determined by the coupled sequence

conditional centers → enumeration order → time of first good leaves → dynamic radius contraction → (8.3)

in addition to the depth distribution of the zero-vector cost.

8.6 Hybrid and oracle ablation results

The five configurations were replayed on 500 transactions per torque condition. All configurations returned the logged globally optimal switching sequence. Table 8.2 reports the total node count.

Table 8.2: Ablation of initialization and search representation. Entries are mean visited nodes; values in parentheses are the 95th percentiles.

Method	10 Nm	50 Nm	90 Nm
Native explicit	451.96 (919)	466.67 (1033)	442.48 (867)
Native embedded	1140.90 (1794)	1179.00 (1845)	1100.20 (1745)
Hybrid	**182.66 (489)**	**191.49 (500)**	**199.92 (504)**
Oracle explicit	160.49 (427)	161.12 (380)	150.97 (313)
Oracle embedded	1123.00 (1746)	1137.90 (1808)	1010.40 (1662)

Relative to the native explicit solver, the hybrid reduces the mean node count by 59.6%, 59.0%, and 54.8% at 10, 50, and 90 Nm, respectively. Relative to the native embedded solver, the corresponding reductions are 84.0%, 83.8%, and 81.8%. Its mean node count lies close to the unattainable oracle explicit baseline, which supplies the true optimum before search begins.

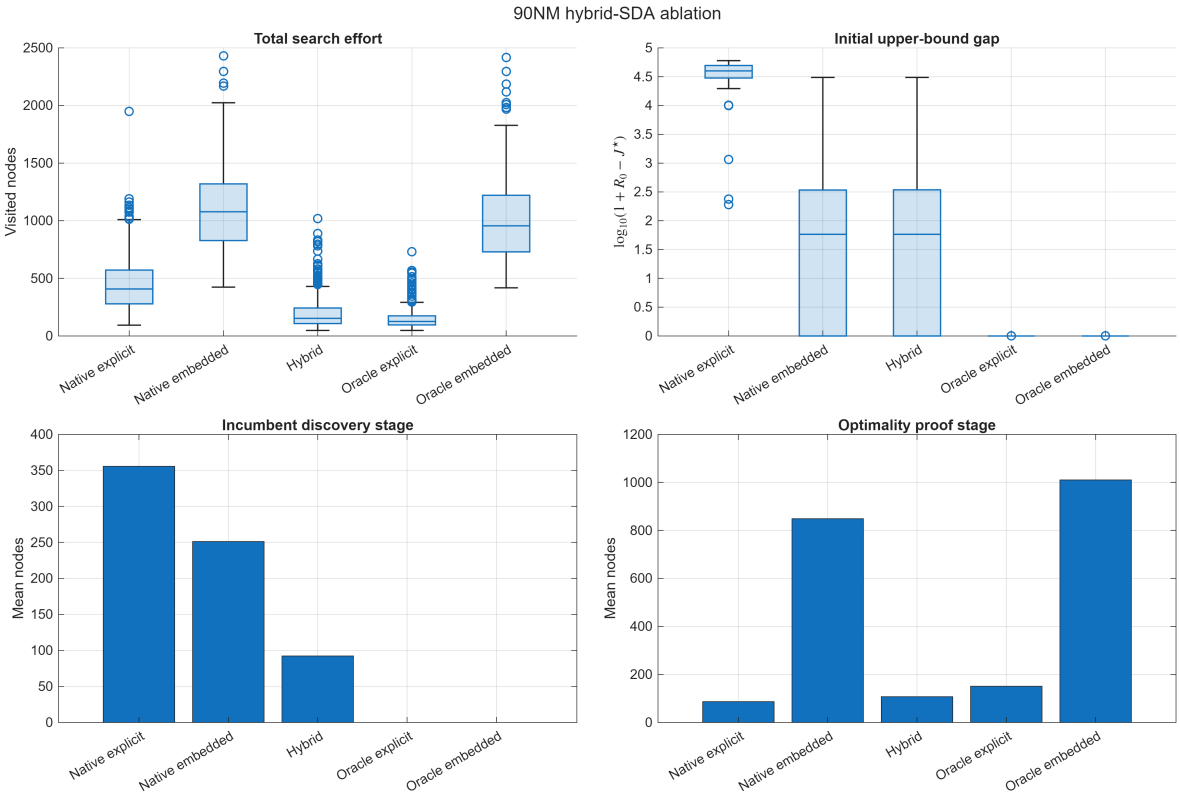


Figure 8.5: Hybrid-SDA ablation at 90 Nm. The upper-left panel compares total visited nodes. The upper-right panel reports the initial upper-bound gap on a logarithmic display. The lower panels partition the trace at the first occurrence of the known optimum. Oracle configurations isolate the search representation by supplying the optimum at node zero.

8.6.1 Evidence for the embedded upper-bound advantage

The native explicit initializer was not optimal in any of the 1500 sampled transactions. Its median initial optimality gap was approximately 49279, 39947, and 39879 at 10, 50, and 90 Nm. The embedded Babai sequence was already optimal in 51.8%, 49.2%, and 40.0% of the respective cases, with median gaps of approximately zero, 2.55, and 57.12. The hybrid inherits exactly this sequence quality because it uses the embedded Babai point and merely reevaluates it under the constant-shift-equivalent explicit objective.

These results support an empirical, not universal, claim: for the investigated machine states, weight, horizon, and variable ordering, embedding the zero-vector objective strongly improves the initial feasible upper bound.

8.6.2 Evidence for the explicit proof advantage

The oracle comparison supplies the optimal sequence to both searches and therefore eliminates initializer quality as an explanation. Even under this condition, the explicit solver requires only 150.97–161.12 mean nodes, whereas the embedded solver requires 1010.40–1137.90. Thus, the explicit search uses approximately 84–86% fewer nodes to certify the same known optimum.

Together with equation (4.9) and the depth-wise pruning peaks, the oracle result provides strong evidence that the explicit formulation offers a more effective branch-and-bound representation. It does not prove universal node dominance for every state or variable ordering, because intermediate non-block depths remain path dependent.

8.6.3 Interpretation of the hybrid improvement

The hybrid combines two complementary properties rather than averaging two controllers. Its embedded Babai step lowers the initial upper-bound gap. Its explicit search retains the block-aligned lower bounds and rapidly certifies or improves that incumbent. The measured reduction is therefore consistent with the intended chain

$$\text{embedded upper bound} \longrightarrow \text{explicit prefix pruning} \longrightarrow \text{lower total search effort.} \tag{8.4}$$

Because the Babai sequence is reevaluated under the explicit objective and the subsequent solver is unchanged, the node reduction does not trade away control optimality.

9 Remaining Analysis

The current results prove leaf equivalence, establish a repeatable complexity difference, and complete the same-prefix comparison of P_i^{exp}, P_i^{emb}, C_i^{eff}, S_i, and initial-sphere membership. They also reject the hypothesis that a larger embedded initial radius is by itself responsible for the additional nodes.

The hybrid and oracle experiments now separate initializer quality from search representation at the node-count level. Before integration into the final thesis, three engineering questions remain. First, execution time rather than node count must be measured, including the additional embedded factor and Babai pass. Second, memory and 10 kHz precomputation cost must be reported. Third, worst-case and high-percentile execution times must be compared with the available real-time deadline.

A further theoretical refinement would compute, for every prefix, the exact best achievable completion cost

$$L_i^{\star}(\boldsymbol{U}_{1:i}) = \min_{\boldsymbol{U}_{i+1:n}} J(\boldsymbol{U}) \tag{9.1}$$

and the lower-bound slack

$$\Gamma_i = L_i^{\star} - P_i. \tag{9.2}$$

The finite 15-bit search space makes this exhaustive offline calculation possible. It would quantify bound tightness independently of traversal order, but it is not required for the already established correctness of the hybrid solver.

The refined causal chain to test is

$$\text{quadratic representation and conditional centers} \Longrightarrow \text{different enumeration order and leaf discovery t} \tag{9.3}$$

The block-boundary penalty and nonuniform prefix offset remain contributing mechanisms, but the evidence no longer supports attributing the observed node difference to the raw initial-radius magnitude alone.

10 Reproducibility Assets

The analysis source, extracted data, validation outputs, representative-case selection, and generated figures are maintained under `Model_2025b/SDA_embedded_analyze`. The production Simulink model and production SDA source are not modified by the profiling workflow.